Master of Business Analytics

Prescriptive Analytics & Optimization

MBAN 5540

Class 04
Duality

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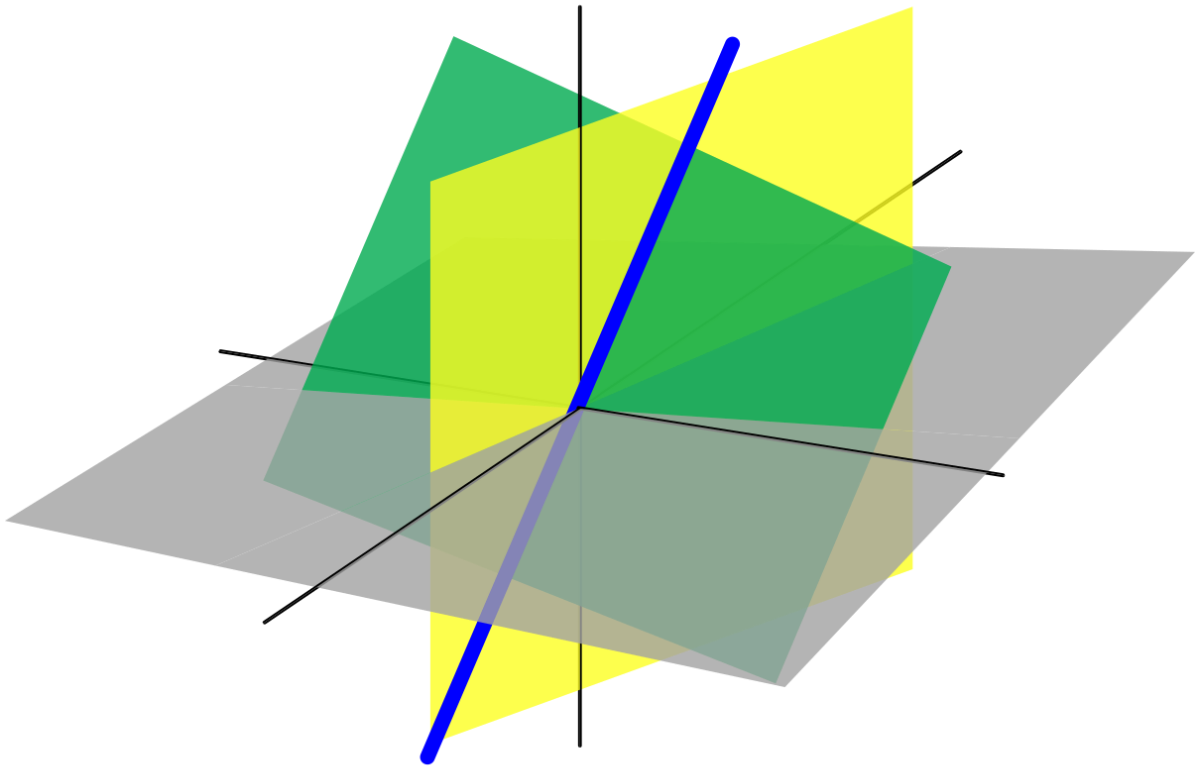
Review

- LP Modeling
- LP Solutions
  - Graphical
  - Excel Solver
  - Gurobi
- LP Sensitivity Analysis

Linear Algebra Review

- We need to review some linear and matrix algebra concepts:
- Vector operations
- Matrix operations



Vector & Matrix

- In our context, a matrix is a 2-dimensional (or tabular) array of numbers, or other objects.
- A matrix is identified by the number of its rows (m) and columns (n)
- For example, a matrix with 2 rows and 3 columns:
- In general, we have

$$A_{2\times 3} = \begin{bmatrix} 34 & 9 & 4 \\ 1 & 72 & 5 \end{bmatrix}$$

Matrix General Form

- In general, we can define a matrix with m rows and n columns as

$$A_{m\times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Matrix Multiplication

- We can multiply matrices only if there is a match between the number of columns of the first one and the number of rows of the second matrix.

- Example: can we add $A_{3\times4}$ by $B_{3\times5}$?

- How about $A_{3\times2}$ by $B_{2\times4}$?

$$\begin{bmatrix} 2 & 7 \\ 1 & 5 \\ 0 & 4 \end{bmatrix} \times \begin{bmatrix} 3 & 0 & 1 & 2 \\ 2 & 1 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 20 & 7 & 2 & 25 \\ 13 & 5 & 1 & 17 \\ 8 & 4 & 0 & 12 \end{bmatrix}$$

Matrix Multiplication

- In general

$$A_{m\times n} \times B_{n\times p} = C_{m\times p}$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{bmatrix}$$

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}$$

Transposed Matrix

- We can switch the rows and columns of a matrix to get its transposed matrix.

$$A_{2\times 3} = \begin{bmatrix} 34 & 9 & 4 \\ 1 & 72 & 5 \end{bmatrix} \rightarrow A^T_{3\times 2} = \begin{bmatrix} 34 & 1 \\ 9 & 72 \\ 4 & 5 \end{bmatrix}$$

- In general form:

$$A_{m\times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \rightarrow A^T_{n\times m} = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}$$

Vector

- A matrix with only one dimension can be called a vector.

$$A_{5\times 1} = \begin{bmatrix} 3 \\ 2 \\ 4 \\ 17 \\ 2 \end{bmatrix}$$

- In general:

$$A_{n\times 1} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \rightarrow A^T_{1\times n} = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}$$

Vector Multiplication

$$A_{1\times n} \times B_{n\times 1} = C_{1\times 1} = \text{a number}$$

$$\begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \times \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 \cdot b_1 + a_2 \cdot b_2 + \cdots + a_n \cdot b_n \end{bmatrix}$$

LP in Matrix Form

- We can use matrices and vectors to represent mathematical models including LP models.

- This is especially necessary when we deal with large scale models with many decision variables and constraints.

- Typically:
  - m is used to show the number of constraints;
  - n is used to show the number of decision variables.

Flair Company Example

$$\begin{aligned} \text{Maximize} \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

where

$$c = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \quad b = \begin{bmatrix} 2400 \\ 1000 \\ 450 \\ -100 \end{bmatrix}, \quad x = \begin{bmatrix} T \\ C \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}$$

LP in Matrix Form

Maximization Model

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

Minimization Model

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax \geq b \\ & x \geq 0 \end{aligned}$$

where

$$c = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

LP in Extended Form

$$
\begin{aligned}
\max \quad & c_1x_1 + c_2x_2 + \cdots + c_nx_n \\
\text{s.t.} \quad & a_{11}x_1 + a_{12}x_2 + \cdots a_{1n}x_n \leq b_1 \\
& a_{21}x_1 + a_{22}x_2 + \cdots a_{2n}x_n \leq b_2 \\
& \vdots \\
& a_{m1}x_1 + a_{m2}x_2 + \cdots a_{mn}x_n \leq b_m \\
& x_1 \geq 0 \\
& x_2 \geq 0 \\
& \vdots \\
& x_n \geq 0
\end{aligned}
$$

$$
\begin{aligned}
\max \quad & \sum_{j=1}^{n} c_j x_j \\
\text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad (i = 1, 2, ..., m) \\
& x_j \geq 0 \quad (j = 1, 2, ..., n)
\end{aligned}
$$

LP in Matrix Form

$$
\begin{aligned}
\max \quad & c^T x \\
\text{s.t.} \quad & Ax \leq b \\
& Dx = d \\
& Sx \geq e \\
& x \geq 0
\end{aligned}
$$

Duality

- For every LP we solve, there is another associated LP (called *Dual*) that we happen to be simultaneously solving.

- It may be used to obtain the solution to the original program (called *primal*), and its variables provide extremely useful information about the set of optimal solutions to the original LP.

- There is exactly one dual variable for each primal constraint.
- There is exactly one dual constraint for each primal variable.

Construct the Dual Problem

Min		Max	
Variable	≥ 0	$\leq$	Constraints
	≤ 0	$\geq$	
	Unrestricted	$=$	
Constraints	$\geq$	≥ 0	Variables
	$\leq$	≤ 0	
	$=$	Unrestricted	

Duality in General

Primal | **Dual**

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned} \quad \longrightarrow \quad \begin{aligned} \min \quad & b^T \pi \\ \text{s.t.} \quad & A^T \pi \geq c \\ & \pi \geq 0 \end{aligned}$$

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax \geq b \\ & x \geq 0 \end{aligned} \quad \longrightarrow \quad \begin{aligned} \max \quad & b^T \pi \\ \text{s.t.} \quad & A^T \pi \leq c \\ & \pi \geq 0 \end{aligned}$$

What happens if we take the dual of dual?

Flair Company Example

Primal

$$\begin{aligned} \max \quad & 7T + 5C \\ \text{s.t.} \quad & 3T + 4C \leq 2,400 \\ & 2T + C \leq 1,000 \\ & \phantom{2T + {}} C \leq 450 \\ & T \qquad \geq 100 \\ & T, C \geq 0 \end{aligned}$$

Primal Standard

$$\begin{aligned} \max \quad & 7T + 5C \\ \text{s.t.} \quad & 3T + 4C \leq 2,400 \\ & 2T + C \leq 1,000 \\ & \phantom{2T + {}} C \leq 450 \\ & -T \qquad \leq -100 \\ & T, C \geq 0 \end{aligned}$$

$$c = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \quad b = \begin{bmatrix} 2400 \\ 1000 \\ 450 \\ -100 \end{bmatrix}, \quad x = \begin{bmatrix} T \\ C \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Dual

$$\begin{aligned} \min \quad & 2,400\pi_1 + 1,000\pi_2 + 450\pi_3 - 100\pi_4 \\ \text{s.t.} \quad & 3\pi_1 + 2\pi_2 + 0\pi_3 - \pi_4 \geq 7 \\ & 4\pi_1 + \pi_2 + \pi_3 + 0\pi_4 \geq 5 \\ & \pi_1, \pi_2, \pi_3, \pi_4 \geq 0 \end{aligned}$$

$$c = \begin{bmatrix} 2400 \\ 1000 \\ 450 \\ -100 \end{bmatrix}, \quad b = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \quad \pi = \begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \end{bmatrix}, \quad A^T = \begin{bmatrix} 3 & 2 & 0 & -1 \\ 4 & 1 & 1 & 0 \end{bmatrix}$$

Example

Primal

$$
\begin{aligned}
\max \quad & 8x_1 + 3x_2 - 2x_3 \\
\text{s.t.} \quad & x_1 - 6x_2 + x_3 \geq 2 \\
& 7x_1 + 7x_2 - 2x_3 = -4 \\
& x_1 \leq 0, x_2 \geq 0, x_3 \text{ is unrestricted}
\end{aligned}
$$

Example

Primal

$$
\begin{aligned}
\max \quad & 8x_1 + 3x_2 - 2x_3 \\
\text{s.t.} \quad & x_1 - 6x_2 + x_3 \geq 2 \\
& 7x_1 + 7x_2 - 2x_3 = -4 \\
& x_1 \leq 0, x_2 \geq 0, x_3 \text{ is unrestricted}
\end{aligned}
$$

Dual

$$
\begin{aligned}
\min \quad & -2\pi_1 - 4\pi_2 \\
\text{s.t.} \quad & -\pi_1 + 7\pi_2 \leq 8 \\
& 6\pi_1 + 7\pi_2 \geq 3 \\
& -\pi_1 - 2\pi_2 = -2 \\
& \pi_1 \geq 0, \pi_2 \text{ is unrestricted}
\end{aligned}
$$

Extended Formulation

Primal

$$
\begin{aligned}
\max \quad & \sum_{j=1}^{n} c_j x_j \\
\text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad (i = 1, 2, ..., m) \\
& x_j \geq 0 \quad (j = 1, 2, ..., n)
\end{aligned}
$$

Dual

$$
\begin{aligned}
\min \quad & \sum_{i=1}^{m} b_i \pi_i \\
\text{s.t.} \quad & \sum_{i=1}^{m} a_{ij} \pi_i \geq c_j \quad (j = 1, 2, ..., n) \\
& \pi_i \geq 0 \quad (i = 1, 2, ..., m)
\end{aligned}
$$

Important Properties of Duality

- The dual problem of dual is primal.

If $x^* = (x_1^*, x_2^*, \dots, x_n^*)$ is the optimal solution of primal;
and $\pi^* = (\pi_1^*, \pi_2^*, \dots, \pi_m^*)$ is the optimal solution of dual, then

$$c^T x^* = b^T \pi^*$$

$$\sum_{j=1}^{n} c_j x_j^* = \sum_{i=1}^{m} b_i \pi_i^*$$

- In fact, π values are shadow prices for the m primal constraints.

Flair Company Example

Primal

$$\begin{aligned} \max \quad & 7T + 5C \\ \text{s.t.} \quad & 3T + 4C \leq 2,400 \\ & 2T + C \leq 1,000 \\ & \phantom{2T + {}} C \leq 450 \\ & -T \leq -100 \\ & T, C \geq 0 \end{aligned}$$

$$x^* = \begin{bmatrix} T \\ C \end{bmatrix} = \begin{bmatrix} 320 \\ 360 \end{bmatrix} \rightarrow c^T x^* = 7(320) + 5(360) = 4,040$$

Dual

$$\begin{aligned} \min \quad & 2,400\pi_1 + 1,000\pi_2 + 450\pi_3 - 100\pi_4 \\ \text{s.t.} \quad & 3\pi_1 + 2\pi_2 + 0\pi_3 - \pi_4 \geq 7 \\ & 4\pi_1 + \pi_2 + \pi_3 + 0\pi_4 \geq 5 \\ & \pi_1, \pi_2, \pi_3, \pi_4 \geq 0 \end{aligned}$$

$$\pi^* = \begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 2.6 \\ 0 \\ 0 \end{bmatrix} \rightarrow b^T \pi^* = 2,400(0.6) + 1,000(2.6) + 450(0) - 100(0) = 4,040$$

Important Properties of Duality

- If the primal (dual) problem is unbounded, then the dual (primal) problem is infeasible.

- If the primal (dual) problem is infeasible, then the dual (primal) problem is either infeasible or unbounded.

Reduced Cost

- Can be calculated by:

$$\mathbf{C} - \mathbf{A}^T \mathbf{\Pi}$$

- In Flair example:

$$c = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \quad b = \begin{bmatrix} 2400 \\ 1000 \\ 450 \\ -100 \end{bmatrix}, \quad x = \begin{bmatrix} T \\ C \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\mathrm{A}^T \pi = \begin{bmatrix} 3 & 2 & 0 & -1 \\ 4 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0.6 \\ 2.6 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix} \qquad c - A^T \pi = \begin{bmatrix} 7 \\ 5 \end{bmatrix} - \begin{bmatrix} 7 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$